

1. Administrative & Study Identification

Full Study Title: A Phase 2, Study for the Treatment of Anemia With Alpha (α)-Thalassemia to Determine the Efficacy and Safety of Luspatercept (BMS-986346/ACE-536) in Adults and Evaluate the Safety and Pharmacokinetics in Adolescents **Short Title/Acronym:** CA056-015 **Protocol Number:** CA056-015 **NCT Number:** NCT05664737 **Posting ID:** 154123056028960064 **Trial Status:** RECRUITING **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Luspatercept (Reblozyl® / BMS-986346/ACE-536) | **ATC Code:** B03XA06 **Phase of Development:** Phase 2 **Medical Monitor:** Bristol Myers Squibb Clinical Operations (clinical.trials@bms.com)

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy and safety of luspatercept plus best supportive care (BSC) versus placebo plus BSC for the treatment of anemia in adult patients with alpha (α)-thalassemia hemoglobin H (HbH) disease, evaluating both non-transfusion-dependent (NTD) and transfusion-dependent (TD) cohorts, and to determine safety and drug levels in adolescent participants. * **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of luspatercept in adolescent participants (aged 12 to <18 years) with α -thalassemia; to assess changes in patient-reported outcomes (NTDT-PRO Tiredness and Weakness domain score), and to measure continuous reductions in red blood cell (RBC) transfusion burden.

2.2 Background & Rationale Alpha-thalassemia hemoglobin H (HbH) disease occurs when ≥ 2 of a patient's 4 α -globin alleles are abnormally expressed, leading to ineffective erythropoiesis and chronic anemia. There is a significant unmet medical need for treatments to alleviate anemia in these patients. Luspatercept promotes late-stage red blood cell maturation and is currently indicated for beta-thalassemia and myelodysplastic syndromes. This registrational Phase 2 trial aims to evaluate its potential as a first-in-class therapeutic option to increase hemoglobin levels and reduce transfusion burden for α -thalassemia, a lifelong disease with limited treatment options.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled (Placebo + Best Supportive Care) * **Blinding:** Double-blind (Adult Phase) / Open-label (Adolescent dose-

confirmation cohort) * **Randomization:** Randomized for adult cohorts

3.2 Planned Enrollment & Duration * **Target N\$:** Approximately 249 participants * **Number of Sites:** ~36 sites per country (Global trial, ex-US focus including China) * **Estimated Study Start:** 09-Dec-2022 * **Estimated Primary Completion:** 16-Jul-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants ≥ 12 years of age with a documented diagnosis of α -thalassemia hemoglobin H (HbH) disease. 2. **Adult Performance Status:** Eastern Cooperative Oncology Group (ECOG) score of 0 or 1. 3. **Adolescent Performance Status:** Karnofsky (age ≥ 16 years) or Lansky (age < 16 years) performance status score ≥ 50 at screening. 4. **Adult TD Cohort:** ≥ 6 RBC units during the 24 weeks prior to randomization. 5. **Adult NTD Cohort:** < 6 RBC units during the 24 weeks prior to randomization, RBC transfusion-free for at least 8 weeks prior to randomization (unless due to acute medical condition), and a mean baseline Hb ≤ 10 g/dL (minimum 2 measurements ≥ 1 week apart within 4 weeks; values within 21 days post-transfusion excluded). 6. **Adolescent TD Cohort:** ≥ 4 RBC events during the 24 weeks prior to enrollment, no transfusion-free period for > 56 days, and a history of regular transfusions for at least 2 years. 7. **Adolescent NTD Cohort:** < 4 RBC events during the 24 weeks prior to enrollment, RBC transfusion-free for at least 8 weeks prior to enrollment, and a mean baseline Hb ≤ 10 g/dL.

4.2 Exclusion Criteria 1. Diagnosis of A-Thal Trait, Hb Bart hydrops, ATRx A-Thal, hemoglobin S/ β -thalassemia, myelodysplasia subtype anemia, or HbE homozygous beta gene mutation. 2. Anemia related to nutritional deficiency, chronic disease, autoimmune hemolytic anemia, or other non- α -thalassemia hemolytic anemias. Hemolysis episodes not related to A-Thal within 8 weeks prior to randomization. 3. History of deep vein thrombosis (DVT), stroke, or other thromboembolic event(s) (except clogged indwelling catheter) requiring medical intervention ≤ 24 weeks prior to randomization. 4. Uncontrolled hypertension. (Controlled defined as BP value corresponding to $\leq$ Grade 1 according to NCI CTCAE Version 5.0). 5. Women who are pregnant, planning to get pregnant during the study, or who are breastfeeding. 6. Prior/Concomitant therapy: Undergone HSCTs or gene therapy (candidates with waiting period of ≥ 12 months are eligible). 7. Use of hydroxyurea treatment ≤ 12 weeks prior to enrollment for NTD participants and ≤ 24 weeks for TD participants. 8. Extramedullary hematopoiesis (EMH) complications requiring treatment to control the growth of EMH mass(es) during the screening period.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Luspatercept recommended dose of 1.0 mg/kg administered once every 21 days (3-week cycles). *** Route of Administration:** Subcutaneous (SC) injection. *** Duration of Treatment:** 48 weeks for the primary evaluation period.

5.2 Concomitant & Prohibited Therapy * Permitted: Best Supportive Care (BSC), RBC transfusions for acute medical conditions. *** Prohibited:** Erythropoiesis-stimulating agents (ESAs) or other investigational therapies during the primary evaluation period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Baseline Hemoglobin measurements (min 2 values ≥ 1 week apart), Medical History, Performance Status
Baseline	Day 1	Randomization, First SC Dose of Luspatercept/Placebo, Quality of Life PRO distribution
Interim	Q3W (Every 21 Days)	Subcutaneous dosing, safety and PK labs, transfusion burden monitoring
Efficacy Eval 1	Weeks 13-24	Continuous 12-week primary efficacy interval for NTD cohort (Hb levels)
Efficacy Eval 2	Weeks 13-48	Continuous 12-week primary efficacy interval for TD cohort (Transfusion burden)
End of Study	Week 48+	Final Vital Signs, Exit Interview, Transfusion Evaluation, Long-term follow up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * Adult NTD Cohort: Number of participants with an increase from baseline of ≥ 1.0 g/dL in mean hemoglobin (Hb) values over a continuous 12-week interval from Week 13 to Week 24 in the absence of RBC transfusions. *** Adult TD Cohort:** Number of participants with $\geq 50\%$ reduction from baseline in RBC transfusion burden (with a reduction of at least 2 units) during any continuous 12 weeks during Week 13-48 compared to the 12-week interval immediately prior to the first dose. *** Adolescent TD and NTD Cohorts:** Dose-limiting toxicities (DLTs) defined as observance of $\geq$ Grade 3-related hemolytic crises or $\geq$ Grade 3-related event outside of the known safety profile occurring within 21 days from their first dose of study therapy.

7.2 Secondary & Exploratory Endpoints * Number of participants with $\geq 33\%$ reduction from baseline in RBC transfusion burden with a reduction of at least 2 units during any continuous 24-week interval on treatment compared to the 24-week interval immediately prior to date of first dose. * The longest duration with $\geq 50\%$ reduction from baseline in RBC transfusion burden

with a reduction of at least 2 units. * Number of RBC transfusion units from Week 1 to Week 48. * Mean change in non-transfusion-dependent beta thalassemia-patient reported outcome (NTDT-PRO) Tiredness and Weakness (TW) domain score. * Safety, tolerability, and pharmacokinetics (PK) in adolescent patients.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection evaluating dose-limiting toxicities (DLTs), thromboembolic events, and changes in blood pressure (specifically Grade 3 to 4 hypertension).
 - **Serious Adverse Events (SAEs):** Reported within 24 hours (including $\geq$ Grade 3 hemolytic crises).
 - **Data Monitoring Committee (DMC):** Standard external/internal safety monitoring overseen by Bristol Myers Squibb.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy endpoints (NTD and TD cohorts evaluated separately); Safety Set for AE/SAE evaluations.
 - **Sample Size Justification:** $N = \sim 249$ designed to adequately power the detection of statistically significant and clinically meaningful increases in hemoglobin (NTD) and decreases in RBC transfusion burden (TD) against placebo.
 - **Handling of Missing Data:** Standard protocol statistical imputation rules for missing Hb lab values and PRO assessments.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Standard clinical site monitoring to verify source data for RBC transfusions and lab results.
 - **Audit Trail:** Fully validated eCRF system for secure data logging and clarification tracking.
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11. Study Identifiers & Governance

- **Primary ID:** CA056-015
- **Registry Number:** NCT05664737
- **Posting ID:** 154123056028960064
- **Sponsor/Lead Organization:** Bristol Myers Squibb
- **Trial Status:** RECRUITING

- **Study Title:** A Study to Determine the Efficacy and Safety of Luspatercept in Adult Participants and to Evaluate the Safety and Pharmacokinetics in and Adolescent Participants With Alpha (α)-Thalassemia
 - **Official Regulatory Title:** A Phase 2, Study for the Treatment of Anemia With Alpha (α)-Thalassemia to Determine the Efficacy and Safety of Luspatercept (BMS-986346/ACE-536) in Adults and Evaluate the Safety and Pharmacokinetics in Adolescents
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12. Clinical Framework (Alpha-Thalassemia Specific)

- **Disease Area:** Anemia
 - **Target Condition:** Anemia in Alpha (α)-Thalassemia (HbH Disease)
 - **Standard Classifications:** MeSH: D000740 | HPO: HP:0001903 | SNOMED: 271737000 | UMLS Preferred Name: Alpha-Thalassemia
 - **Diagnosis Threshold:** Confirmed diagnosis of HbH disease. Mean baseline Hb ≤ 10 g/dL for NTD cohorts.
 - **Medical Methodology:** Luspatercept (ATC Code B03XA06) combined with Best Supportive Care.
 - **Optimization Target:** Erythropoiesis stimulation to achieve transfusion independence or a $\geq 50\%$ reduction in RBC burden.
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13. Study Procedures & Methodology

- **Management Pathway:** Luspatercept SC injections every 21 days + standardized RBC transfusion threshold guidance.
 - **Education Protocol (HCP):** Site initiation visits covering SC administration, AE monitoring (hypertension, thrombosis), and strict transfusion event logging.
 - **Education Protocol (Patient):** Informed consent, instructions on completing the NTDT-PRO Tiredness and Weakness daily/weekly diary.
 - **Quality Audit Mechanism:** Transfusion event source-document verification and central laboratory validation.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Primary evaluation up to 48 weeks, extending into long-term follow-up.
 - **Onsite Visit Cadence:** Every 3 weeks (Day 1 of each 21-day cycle) for study drug administration and safety labs.
 - **Remote Monitoring Frequency:** Inter-cycle as clinically indicated.
 - **Checklist Review Interval:** Every 12 weeks for continuous primary endpoint interval evaluations.
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15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					